

Isaac Jesán Velázquez Reséndiz

Nationality: Mexican

Academic position: Doctoral Researcher at [IMT Atlantique](#) - Brest, France

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Academic profile

Physicist and renewable energy engineer working on the intersection of magnonics and neuromorphic computing. My research aims to develop magnetic-material-based technologies to reduce energy consumption in electronics and contribute to a more sustainable technological future. I am committed to the advancement of low-energy quantum technologies through theoretical and experimental research, while actively engaging in teaching, collaboration, and scientific dissemination.

Scientific interests

- Open quantum systems
- Magnonics
- Light-matter interactions
- Neuromorphic computing
- Quantum field theory
- Quantum information theory

Studies

- **IMT Atlantique - Brest, France**
PhD student, Dr. Vincent Vlaminck. May 2026 - Present
- **Physics Institute, UNAM, Mexico**
Master on physics (9.84/10), Dr. Yuri G. Rubo, Honors distinction, August 2023 - August 2025
- **Renewable Energies Institute, UNAM, Mexico**
Renewable energy engineering (9.25/10), Dr. Yuri G. Rubo, Honors distinction, 2018 - 2023
- **Jaen University, Spain**
Exchange semester, January - June 2022
- **Quilmes National University, Argentina**
Research internship, Dr. Santiago Manuel Garrido's group, June - August 2019

Articles

- César Benito, **I. Jesán Velázquez-Reséndiz**, Alejandro Bermudez, Optimizing bias-tailored quantum error correction beyond code-capacity noise *arXiv preprint arXiv:2606.17709 (2026)*
- **I. Jesán Velázquez-Reséndiz** and Yuri G. Rubo, Polarization dynamics of trapped polariton condensates with $\mathcal{PT}$ -symmetry *Phys. Rev. B 109, 085312 (2024)*

Distinctions and Funding

- Master's scholarship: CONAHCYT (August 2023 - September 2025)
- Quantum Excellence Certificate.- IBM Quantum, 2024 Qiskit Global Summer School: The Path to Utility
- Quantum Excellence Certificate.- IBM, Quantum, 2022 Qiskit Global Summer School: Quantum Machine Learning
- Quantum Excellence Certificate.- IBM Quantum, 2021 Qiskit Global Summer School: Quantum Simulation

Experience

- **PEPR SPIN - SWING, ANR-22-EXSP-0004 (IMT Atlantique - Brest)**: Spin Waves for advanced signal processing, Dr. Vincent Vlamincq, *Research assistant: Magnon Interferometry with Topological Spin Textures*, (May 2026 - Present)
- **MODULARIS, OPE03595 (IFT UAM-CSIC) - MILLENION Quantum**: Modular Logical Qubits, Dr. Alejandro Bermúdez Carballo, *Research assistant: Development of fault-tolerant logical qubits in a trapped-ion architecture using surface-code-based protocols and AI-based decoder*, (September 2025 - April 2026)
- **PAPIIT IN108524 (UNAM)**: Quantum theory of polaritonic condensation and its applications, Dr. Yuri G. Rubo, *Research assistant: Theoretical and numerical analysis of the quantum properties of a polaritonic condensate through the Lindblad master equation formalism*, (2023 - 2025)
- **PAPIIT 106320 (UNAM)**: Polaritonic condensates for quantum simulation and computing, Dr. Yuri G. Rubo, *Research assistant: Theoretical analysis of the dynamics of a polaritonic Bose-Hubbard dimer with $\mathcal{PT}$ -symmetry*, (2020 - 2022)
- **PAPIME PE110319 (UNAM)**: Prototype design and Manual of experimental activities for the LIER, Dr. Jorge Alejandro Wong Loya, *Research assistant: Construction and calibration of over 50 type T thermocouples and writing laboratory manuals for permeability and heat transfer experiments*, (2019 - 2021)
- **CUAM Morelos**: Research Congress CUAM-AcMor, *Reviewer of high school projects with technological and social impact in the community of Morelos*, (May 2019)
- **Renewable Energies Institute, UNAM**: National Congress of Renewable Energies Students, *Staff*, (November 2018)

Teaching

- **Electricity and Magnetism**
 - *Teaching assistant: Renewable energy engineering, 2025-2*
- **Classical Electrodynamics Propaedeutic**
 - *Teaching assistant: Admission to the Postgraduate Degree in Physical Sciences, 2024-2*
- **Electricity and Magnetism**
 - *Teaching assistant: Renewable energy engineering, 2024-1*
- **Electricity and Magnetism**
 - *Teaching assistant: Renewable energy engineering, 2023-2*

Conferences and Posters

- Conferences
 - Etic Tlahuilli: Polariton School, *Polarization dynamics of trapped polariton condensates with $\mathcal{PT}$ – symmetry*, Physics Institute, UNAM, Mexico, June 2024
 - Morelos Science Student Seminar, *The rise of quantum computing*, Institute of Physical Sciences, UNAM, Mexico, April 2023
 - Thesis in short, *Quantum to the rescue*, Renewable Energies Institute, UNAM, Mexico, October 2022
- Posters
 - National Physics Congress, Mexican Physics Society, Mexico, October 2024
 - XIX PCF-UNAM Student Congress, Physics Institute, UNAM, Mexico, December 2023
 - Etic Tlahuilli: Polariton Meeting, Physics Institute, UNAM, Mexico, November 2023

Languages

- Spanish: Native
- English: Advanced (C1-TOEFL)
- French: Intermediate (B1)

Technical skills

- Numerical simulations of open quantum systems and realistic noise models
- Quantum error correction and fault-tolerant quantum computing (surface codes, decoders)
- Scientific programming in Python (numpy, qutip, qiskit, stim, pytorch) and Mathematica
- Collaborative development using Git and GitHub

Science dissemination

- The money that does not exist: Cryptocurrencies and renewable energies (in Spanish)

Soft skills

- Team work
- Persistence and perseverance
- Adaptability
- Fast learner and autodidact
- Honest
- Solidarity
- Passionate
- Respectful